

Expt 5	Design of High Pass Filter using Windowing Techniques
Date:	5.2. Generate the coefficients of an FIR High-Pass filter using Hamming windowing techniques

CODE :-

```

clc; clear; close all;

%% --- Input Parameters ---

fp = 1000; % High-pass cutoff frequency (Hz)
fs = 8000; % Sampling frequency (Hz)
N = 50; % Filter order
wc = fp/(fs/2); % Normalized cutoff (0–1)

%% --- FIR High Pass using Hamming Window ---

b_hamm = fir1(N, wc, 'high', hamming(N+1));

disp('--- High-Pass Coefficients (Hamming Window) ---');
disp(b_hamm);

%% --- Frequency Response Plots ---

figure;
freqz(b_hamm,1);
title('High-pass FIR using Hamming Window');
title('High-pass FIR using Hanning Window');

```

OUTPUT :-

--- High-Pass Coefficients (Hamming Window) ---

-0.0007

0

0.0009

0.0016

0.0015

0

-0.0024

-0.0044

-0.0039

0

0.0060

0.0103

0.0088

0

-0.0128

-0.0217

-0.0184

0

0.0268

0.0464

0.0410

0

-0.0725

-0.1567

-0.2240

0.7493

-0.2240

-0.1567

-0.0725

0

0.0410

0.0464

0.0268

0

-0.0184

-0.0217
-0.0128
0
0.0088
0.0103
0.0060
0
-0.0039
-0.0044
-0.0024
0
0.0015
0.0016
0.0009
0
-0.0007

